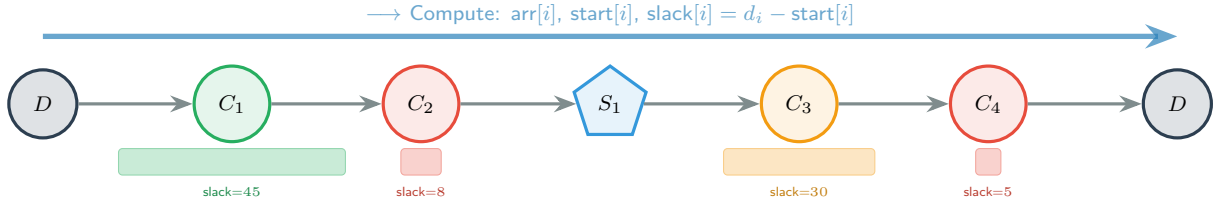


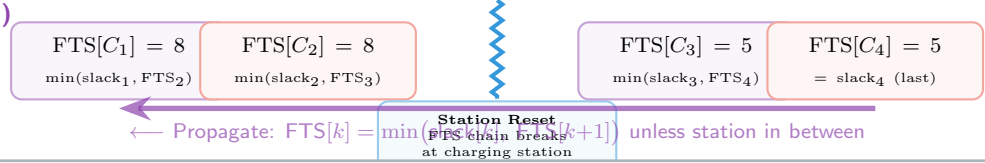
TimeSlackDestroy Operator

Forward-Backward Time Slack Analysis with Liberation Score

Phase 1: Forward Pass



Phase 2: Backward Pass (FTS)



Liberation Score for customer C_i :

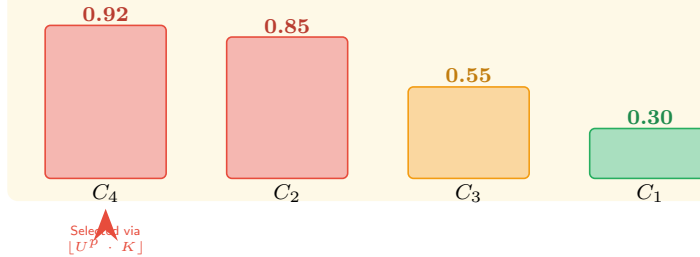
$$L(C_i) = \underbrace{\alpha \cdot \frac{1}{FTS[i] + \varepsilon} \cdot (1 + \text{lockCount})}_{\text{(A) FTS Criticality}} + \underbrace{\beta \cdot \Delta t_{\text{saved}}}_{\text{(B) Actual Time Saved}} + \underbrace{\gamma \cdot w_i}_{\text{(C) Wait at } C_i}$$

(A) Low FTS $\Rightarrow$ tight bottleneck.
lockCount = # predecessors whose FTS is lim

(B) Time freed by bypassing C_i : $\Delta t = t_{\text{curr}} - t_{\text{new}}$.
Shortcutting the detour.

(C) Large wait $\Rightarrow C_i$ arrived too early, blocking the route.
Removing it improves flow.

Top- K Candidates (sorted by Liberation Score)



Biased random:
 $\text{idx} = \lfloor r^p \cdot K \rfloor$
 $r \sim \text{Uniform}(0, 1)$
 p = exploration factor
 $p \uparrow \Rightarrow$ more greedy
 $p \downarrow \Rightarrow$ more random

Phase 4: Biased Selection

Lock Count — FTS Criticality Amplifier



Backward scan from C_i :
• C_c : $FTS=5 = FTS[C_i] \Rightarrow \text{lock}++$
• C_b : $FTS=5 = FTS[C_i] \Rightarrow \text{lock}++$
• C_a : $FTS=5$ but check stops (no station in between)
 $\Rightarrow$ Removing C_i liberates 2 predecessors from tight FTS

lockCount = 2
(same FTS $\Rightarrow C_i$ is the bottleneck)